

Experiment No.:8

Case Study: Healthcare Monitoring System Using Python (Google Colab)

Aim

To develop and simulate a Healthcare Monitoring System using Python in Google Colab that continuously monitors patients' vital signs, identifies abnormal health conditions, displays real-time graphical visualization, and generates a final healthcare report.

Objective

The objective of this experiment is to simulate a Healthcare Monitoring System using Python in Google Colab. The system continuously generates and monitors the body temperature, heart rate, and blood oxygen saturation (SpO₂) of 10 patients using simulated data. Each patient's health parameters are displayed in real time along with graphical visualizations. The system compares the generated values with predefined medical threshold values and classifies the patient's condition as either **NORMAL** or **EMERGENCY**. After monitoring all patients, a final healthcare report is generated and saved as a CSV file. This experiment demonstrates real-time patient monitoring, automated health status classification, data visualization, and report generation in a simulated healthcare environment.

Software Requirements

- Google Colab
- Python 3.x

Python Libraries Used

- pandas
- matplotlib
- random
- time
- IPython.display

```
# =====
```

```
# Case Study : Healthcare Monitoring System
```

```
# Google Colab Simulation
```

```
# =====
```

```
import random
```

```
import time
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
from IPython.display import clear_output
```

```
# -----
```

```
# Patient Names
```

```
# -----
```

```
patients = [
```

```
    "Patient 1","Patient 2","Patient 3","Patient 4","Patient 5",
```

```
    "Patient 6","Patient 7","Patient 8","Patient 9","Patient 10"
```

```
]
```

```
temperature_data=[]
```

```
heart_data=[]
```

```
spo2_data=[]
```

```
# -----
```

```
# Simulation
```

```
# -----
```

```
for patient in patients:
```

```
    temp=round(random.uniform(35.5,40.5),1)
```

```
    heart=random.randint(55,130)
```

```
    spo2=random.randint(88,100)
```

```
    temperature_data.append(temp)
```

```
    heart_data.append(heart)
```

```
    spo2_data.append(spo2)
```

```
    if temp>38 or heart>110 or spo2<92:
```

```
        status="EMERGENCY"
```

```
        color="red"
```

```
    else:
```

```
        status="NORMAL"
```

```
        color="green"
```

```
    clear_output(wait=True)
```

```
    print("="*60)
```

```
    print("HEALTHCARE MONITORING SYSTEM")
```

```
    print("="*60)
```

```
    print("Patient :",patient)
```

```
    print("Temperature :",temp,"°C")
```

```
print("Heart Rate :",heart,"BPM")
print("SpO2 :",spo2,"%")
print("Status :",status)

plt.figure(figsize=(12,4))

# Temperature
plt.subplot(1,3,1)
plt.bar(["Temp"],[temp],color="orange")
plt.ylim(30,45)
plt.title("Temperature")

# Heart Rate
plt.subplot(1,3,2)
plt.bar(["Heart"],[heart],color="blue")
plt.ylim(40,150)
plt.title("Heart Rate")

# SpO2
plt.subplot(1,3,3)
plt.bar(["SpO2"],[spo2],color="color")
plt.ylim(80,100)
plt.title("SpO2")

plt.tight_layout()
plt.show()
```

```
time.sleep(2)

# -----
# Final Report
# -----

report=pd.DataFrame({

    "Patient":patients,
    "Temperature":temperature_data,
    "Heart Rate":heart_data,
    "SpO2":spo2_data

})

print("\nFINAL REPORT\n")
print(report)

report.to_csv("Healthcare_Report.csv",index=False)

print("\nCSV File Saved Successfully")
```

Theory

A Healthcare Monitoring System continuously monitors patients' vital signs to assess their health condition. In hospitals, medical sensors measure body temperature, heart rate, and blood oxygen saturation (SpO₂), allowing doctors to detect emergencies quickly.

In this experiment, patient data are simulated using randomly generated values instead of real medical sensors. The system monitors one patient at a time and compares each patient's vital signs with predefined threshold values. If any value exceeds the normal range, the patient is classified as **EMERGENCY**; otherwise, the patient is classified as **NORMAL**.

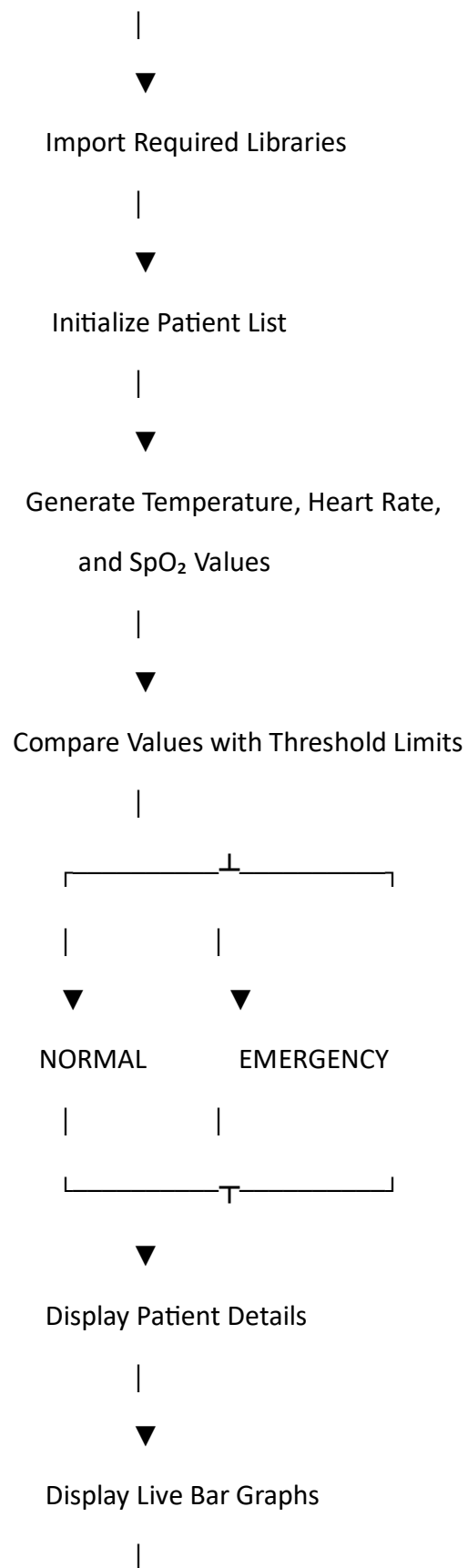
The patient's vital signs are displayed through real-time bar graphs, making it easier to monitor health conditions visually. At the end of the simulation, a complete healthcare report is generated and saved as a CSV file for future analysis.

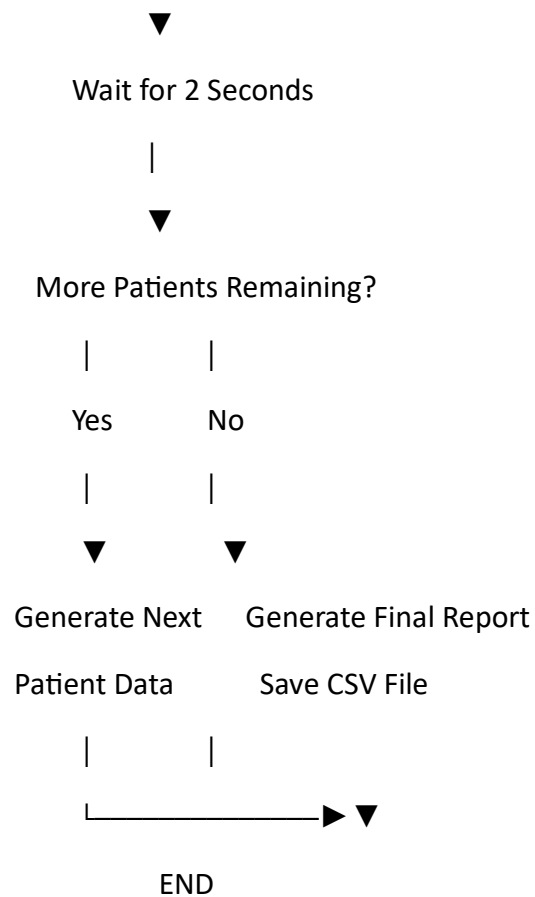
Algorithm

1. Start the program.
2. Import all required Python libraries.
3. Create a list of 10 patients.
4. Initialize empty lists to store patient data.
5. Generate random Temperature, Heart Rate, and SpO₂ values.
6. Compare the values with predefined threshold limits.
7. Classify the patient as **NORMAL** or **EMERGENCY**.
8. Display patient details.
9. Display real-time graphs for Temperature, Heart Rate, and SpO₂.
10. Wait for 2 seconds.
11. Repeat until all patients are monitored.
12. Generate the final healthcare report.
13. Save the report as a CSV file.
14. Display "CSV File Saved Successfully".
15. Stop the program.

Flowchart

START





Threshold Values Used

Parameter	Normal Range	Emergency Condition
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Temperature	36.5°C – 37.5°C	>38°C
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Heart Rate	60–100 BPM	>110 BPM
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SpO ₂	95–100%	<92%
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Program Description

The Python program simulates a Healthcare Monitoring System for 10 patients. For each patient, the system randomly generates body temperature, heart rate, and SpO₂ values. These values are compared with predefined threshold values to determine whether the patient's condition is **NORMAL** or **EMERGENCY**. The patient's health details are displayed on the screen, and three bar graphs visualize the temperature, heart rate, and oxygen saturation levels. The

monitoring process continues until all patients are evaluated. Finally, the patient data are stored in a pandas DataFrame, displayed as a report, and saved as a CSV file named **Healthcare_Report.csv**.

Input

- Total Patients = 10
- Simulated Temperature
- Simulated Heart Rate
- Simulated SpO₂

Output

- Patient Details
- Health Status (NORMAL/EMERGENCY)
- Temperature Graph
- Heart Rate Graph
- SpO₂ Graph
- Final Healthcare Report
- CSV Report File

Sample Output

Patient	Temperature	Heart Rate	SpO ₂	Status
Patient 1	36.8°C	78 BPM	98%	NORMAL
Patient 2	39.2°C	118 BPM	90%	EMERGENCY
Patient 3	37.1°C	82 BPM	97%	NORMAL
Patient 4	38.5°C	112 BPM	91%	EMERGENCY

Patient Temperature Heart Rate SpO₂ Status

Patient 5 36.7°C 76 BPM 99% NORMAL

Observations

- Patient vital signs are generated automatically.
- The system monitors one patient at a time.
- Temperature, Heart Rate, and SpO₂ are displayed using live graphs.
- Patients exceeding threshold limits are classified as **EMERGENCY**.
- A final healthcare report is generated.
- The report is successfully saved as a CSV file.

Applications

- Smart Hospitals
- Patient Monitoring Systems
- Intensive Care Units (ICU)
- Telemedicine
- Remote Healthcare Monitoring
- Medical Education
- Healthcare Data Analysis
- IoT-Based Healthcare Systems

Advantages

- Real-time patient monitoring.
- Automatic emergency detection.
- Easy-to-understand graphical visualization.
- Generates healthcare reports automatically.

- Saves patient data as a CSV file.
- Cost-effective educational simulation.
- Can be integrated with IoT medical sensors.
- Suitable for academic and research purposes.

Limitations

- Uses simulated patient data.
- No real medical sensors are connected.
- Does not send alerts to doctors.
- Suitable only for educational demonstration.
- Internet connectivity is not required.

Result

The **Healthcare Monitoring System** was successfully simulated using Python in Google Colab. The system monitored the vital signs of 10 simulated patients by generating body temperature, heart rate, and SpO₂ values. It accurately classified each patient's condition as **NORMAL** or **EMERGENCY**, displayed real-time graphical visualizations, and generated a final healthcare report that was saved as a CSV file. Thus, the experiment successfully demonstrated real-time patient monitoring, automated health status classification, graphical visualization, and report generation in a simulated healthcare environment.

HEALTHCARE MONITORING SYSTEM

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Patient : Patient 10

Temperature : 37.9 °C

Heart Rate : 107 BPM

SpO₂ : 95 %

Status : NORMAL

